

Social Distancing at Bars

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1 Introduction

We are living in a time where the Coronavirus (COVID-19) pandemic is causing a large impact on our social lives. To counteract the effect of Corona, this paper proposes an AI (Artificial Intelligence) driven solution allowing for a safer environment in entertainment locations, in particular bars and cafes. The main research question we focus on is: “Is it possible to increase people’s compliance with social distancing rules in bars through the introduction of an AI-driven wristband solution?”

To be able to answer the main research question and propose an effective solution, there are three fields of research that need to be covered. First of all, how does one increase people’s compliance? Second, what social distancing rules are set in place in bars, how are these rules enforced currently and how can this be improved? Lastly, what would a wristband solution look like and which AI systems can be incorporated?

Further evaluation is necessary to validate our proposed solution, for which we will set up an evaluation plan. Nevertheless, we hope that our idea can contribute to both the general social health of people and allow for effective combating the Coronavirus.

2 Problem Statement

The ongoing pandemic has compelled governments to implement several rules, with which come a variety of lifestyle changes. In short, the rules prompt citizens to stay at home as much as possible and when having to go out, to stay a minimum of 1,5 meters away from other people. These rules apply to everyone over the age of 18 in all public places, with exception of household members. [6] As time has gone by, the government has become more lenient on the stay-at-home rule and public places have slowly opened again.

Bars were one of the public places greatly affected by the initial lockdown. They solely rely on visiting customers, so an indefinite lockdown is no longer possible as the need to support the economy is surfacing rapidly. To make sure that these bars can stay open, there need to be sufficient and most importantly effective reminders. Unfortunately, when reopening, these bars are struggling with implementing measurements given by the government. This was for example the case in Café ‘t Pötterke in Maastricht, where the bar had to close its doors for two weeks due to customers ignoring the social distancing rules. [11] An example of the measurements given were requiring all staff to wear cloth face coverings and posting signs in highly visible locations that promote everyday protective measures. [4] As it turns out, these measurements were not enough.

Apart from an economic perspective, the closing of bars has had a significant impact on the mental health of people, specifically students. The reason for this is that students’ main source of interpersonal communication is in informal settings, a popular one being bars. [2] With the closing of bars, students have less opportunity to spontaneously socialize with others, leading to isolation and thus a worsening of their mental health.

To sum it up, there are both economic and mental health reasons to have bars stay open to students. To be able to do this safely, more effective measurements have to be implemented in the bars. This paper takes up this unique case and proposes a solution using modern technology and more specifically AI (Artificial Intelligence).

3 Literature Research

To be able to come up with a working solution, one must first understand the problem better. Questions that need answering are: what is COVID-19, where did it originate and how does it spread? Furthermore, is the implemented social distancing rule effective and does it have any implications? On top of that, what solutions have been introduced or are being researched?

3.1 COVID-19: Origin and Transmission

A cluster of pneumonia cases occurred in Wuhan, China in December 2019. This was caused by a then recently identified beta-coronavirus, later to be named COVID-19. The disease was fast-growing, claiming 2.873 deaths out of a total 79.968 cases in mainland China as of March 2020. [17] As for the virus's origin, its genome is 96,2% identical to that of a bat. After further evaluation and analyses, researchers confirmed the notion that bats are the natural hosts of the virus origin. There is a possibility that the virus was transmitted via an unknown intermediate host to humans. [18]

The way COVID-19 spread further is from person to person. This was confirmed by a study researching the clinical characteristics of the virus throughout January 2020. The study extracted data regarding 1.099 patients spread over 552 hospitals in mainland China. They found that among the non-residents of Wuhan, 72,3% had been in physical contact with at least one Wuhan resident, including 31,3% that visited the city themselves. This is most likely where they contracted the virus. [7] Someone who is ill from the virus produces millions of copies of the virus inside their body, which are then mainly located in the lungs, throat and nasal cavity. When talking, coughing or sneezing, lots of tiny droplets are pushed out of the lungs, throat and nasal cavity. These droplets carry the virus and can, upon inhalation or touch, infect others. The droplets that are the main cause of infection are somewhat larger ones, but they rarely travel beyond 1,5 meters, thus why people need to keep this distance away from each other. Moreover, in late September 2020, it has been announced that the reproduction number has risen to 1,17. Meaning that 100 COVID-19 infected people will collectively infect 117 more. [14]

3.2 Effectiveness of Social Distancing

There are two types of social distancing: staying quarantined at home and staying 1,5 away from others. When ordering citizens to stay home, it is unrealistic to assume that everyone will without exceptions, stay home. A study conducted in September 2020 found consistently that quarantine is indeed important in reducing incidence and mortality, although there still is uncertainty over the effect's magnitude. There need to be other public health measures implemented, such as the social distancing rule of 1,5 meters, to ensure the effectiveness of quarantine itself. A balance of these measurements can be maintained by monitoring the outbreak and the implemented measures' effect. [12]

Whether social distancing is essential and effective amid the pandemic is further explained in a study observing the growth rate of COVID-19 in the US during March and April 2020. Additionally, this study compared the observed rates to the predicted rates. It was found that the growth rates started decreasing eight days after the earliest social distancing measures were set in place. The mean exponential growth rate over the full period was 16,2% without any interventions. The social distancing measures would have decreased the growth rate to 11%. In reality, the actual number fell to 3%, as a result of both social distancing measures and shelter-in-place orders being set in motion. [5] From this information, it can be concluded that keeping a distance of 1,5 meters is important to control the virus, but this does not come without problems.

3.3 Implications of Social Distancing

After roughly six months of social distancing and staying home, while effective, the latter is causing several mental health implications.

A study that has been tracking the mental well-being of several age groups from 2009 until recently in the UK was able to clearly show the difference over time. This study included 36.000 different individuals, with an age range from 16 to 65+ and different genders. These individuals were asked to fill in a survey every 1-2 years, answering several questions in regards to their mental well-being. This survey was done using a Likert scale from 1 to 4, where a higher score indicated poorer mental health. The age group from 16 - 24 showed a significant difference in well-being before and during the pandemic. The scores for feeling unhappy or depressed and general happiness jumped from an average score of 1,9 in the span of January 2009 to May 2019 to a score of almost 2,4 during the pandemic. The difference was even greater for enjoying day-to-day activities, where over the years a steady score between 2,0 and 2,1 jumped to a solid 2,5. When these statistics were broken down into two genders, male and female, females scored on average 0,2 higher in both categories, indicating that they had poorer mental well-being compared to their male counterparts. While this particular study did not ask the participants the reasons for the change in their mental well-being, the collective difference indicated a collective cause. The change of a pandemic happening impacted all of the participants' lives without exceptions and thus it can be safely said that the pandemic was the reason for these results. [1] To conclude, the pandemic had a significant negative impact on the mental health of those in the age group 16 - 24, which includes students.

One of the reasons for the decline in mental health includes detachment from loved ones, especially those that are in quarantine and are then quick to feel frustrated, bored and potentially depressed. There is also a risk of anxiety, a result of fearing getting sick and unclear social distancing guidelines. [16] The effects are seen in about 25 percent of college students following the COVID-19 outbreak, mainly because quarantine gradually distances people from each other. This causes an absence of interpersonal communication, worsening depression and anxiety. [3] To combat this, there needs to be a way of being able to come together and socialize, in this paper's particular case in bars, while considering

the safety regulations. This also, to prevent long-lasting effects of the pandemic on students' mental health.

3.4 Current Research and Existing Solutions

Throughout 2020 many proposals have been done to enforce social distancing using modern technology. These proposals include extensive research but as far as implementation to the general public goes, there is not much, if anything, to find.

Surveillance cameras. A common tool in proposals for enforcing social distancing are surveillance cameras. In one proposal, a camera tracks individuals in a set space and generates a bounding box for each individual in real-time. This system makes use of a deep learning framework, an implementation of AI, to automate the object detection and tracking process. While not used in this proposal, it can specifically be done with computer vision, a tool that can detect individuals and detect whether they comply with the social distancing rules. Additionally, computer vision can be trained so that it is able to identify the usage of masks or potential symptoms. [15] The aforementioned bounding boxes aid in identifying the distance between individuals and groups using a vectorized approach. [13]

An issue with this approach is that it is sensitive to the spatial location of the camera. One way to better this is by either having the system take the viewing angle into account, or having multiple cameras set up in different locations, giving a better view. Another problem when using surveillance cameras is that it is unable to recognize individuals from the same household, as that is an exception to the social distancing rule. This problem could be avoided using computer vision, but apart from facial recognition being more difficult when individuals are wearing masks, the practical implementation brings even more problems. When using this in a bar setting, there needs to be a way of inputting the data of individuals in the system, this includes registering their face and information on who their housemates are. Besides it being impractical, it might bring up privacy issues.

Wearables. Another common tool to enforce social distancing are wearables. A big upside of these is that users can stay completely anonymous, which already makes for a more appealing solution compared to the cameras.

Ultrasound or ultrasonic positioning system (UPS) is a way of tracking wearables on individuals that have ultrasonic beacons (UBs). Beacon units periodically broadcast ultrasonic and radio frequency (RF) messages together with their unique ID numbers. Based on these, the receiver's position can be determined by using calculation methods such as trilateration or triangulation. Positively, the UPS does not need a line of sight between the transmitter and receiver, and it does not interfere with electromagnetic waves. [8] Unfortunately, the propagation of ultrasound waves is limited, limiting the use to indoor environments

only. This does not present a problem for the current setting of an indoor bar. However, the system reaches a limit when a large number of wearables is present due to its high complexity. [10]

Bluetooth technology is another solution, which works in both indoor and outdoor environments. It includes all the good features that the UPS has, but without a limit of wearables. Bluetooth is also a more desirable solution compared to Wi-Fi localization, as Bluetooth can be implemented in smaller wearables and the beacons of the system are easier to deploy as it is powered by a battery. With Bluetooth, there are two options, either using existing devices such as phones, or creating a new type of wearable. Here, the latter is a better option, as it avoids privacy issues and other incidents such as an empty battery halfway through the day or accidentally switching off the Bluetooth feature. [9]

When designing a solution to be used in bars, it should be as simple as possible. It should be easy for users, here bar customers and staff, to understand and easy for a bar owner to set up. From the found literature it can be concluded that a wearable is the best option to distribute in bars, as it is easy for the user to keep on their person. A Bluetooth device also seems to be the best option when considering the set up as there are only two components: the wearables that are given to the customers and the central device, that can display information.

4 Knowledge Acquisition

Introducing a technical solution to a social situation requires a good understanding of human behaviour and exploration of expected outcomes when humans are interacting with a machine. In order to achieve a general understanding, we interviewed two people who specialized in Psychology. Dr. Mieke Donk, associate professor at the VU and Drs. M.M. Andalusi, a psychologist with education in CBT (Cognitive Behavioural Therapy). The full interviews can be found in the appendix.

The basic concept of introducing bracelets to people socializing at a bar to monitor and steer social distancing is explained to the interviewees. When asked for a first opinion on this concept Drs. Andalusi acknowledges the difficulty of upholding social distancing rules stating that “People want more freedom but as a result, there is more risk. Especially in bigger social environments”. Introducing a device that keeps people conscious of the distancing rules can help reduce this risk, especially in situations where alcohol is involved.

When imposing a rule on people to wear a bracelet, it can be helpful to present it as an access key to something people want, that is ‘to socialize with other people’. Similarities can be seen with festival bands in it taking the same physical effort. However, the ongoing pandemic could create more negative thoughts. Less freedom, more monitoring.

The involvement of alcohol might influence the behaviour of people and their responsiveness to the bracelets. Advice from Dr. Donk is to make use of a clear and annoying signal when people are not committing to the social distance. Also, Dr. Donk points out that a locking system on the bracelet might help

prevent people from taking off the bracelet when under influence. Introducing the bracelet at the front door and unlocking it again when leaving. This also matches well with the concept of introducing the bracelet as an access key to socialize with other people.

Drs. Andalusi adds to this by stating that it would be a good addition to have the system monitor if a person is showing ‘behaviour under influence’, combined with clear consequences. When further discussing possible consequences for people’s behaviour in general, Drs. Andalusi explores the possibility of either positive or negative consequences. Rewarding people for good behaviour by offering discounts or free consumables and punishment by giving out fines or denying access to the bar for a certain period of time. Drs. Andalusi emphasises that positive feedback might be a socially preferred option, promoting healthy behaviour in a time where people who do keep to the rules are receiving no feedback. Although negative feedback might be difficult to refrain from when people are breaking important rules, especially when people are under influence.

5 Proposed Solution

This paper is an attempt to introduce an AI rules-based system that helps regulate distancing at bars through the use of a bracelet. When people enter the bar, they will all receive a special electronic bracelet to wear for the entire duration of the party. The bracelet has a specific chip with a unique chip ID and once activated, it is directly connected to the system using Bluetooth 4.0 protocol. This system solution is designed to keep track of the location proximity of all bracelets. The system’s interface is presented as an app, to communicate with a staff member monitoring the activity in the bar, here referred to as the manager’s display.

Optionally, the bracelet’s unique chip ID can be linked with the wearer’s contact information, such as their phone number. When a few days after a person’s visit it is discovered that they were infected with COVID-19, the system can contact all the other people who were near the infected person immediately. However, as this may raise some privacy concerns, this idea will not be further elaborated in this paper’s solution.

Grouping bracelets allows for two or more devices to get close without trouble, which is useful for people from the same household. When two or more non-grouped devices get close to each other (nearing 2m), a warning in the form of a visual light flash is set off. When the people miss it and move even closer to each other and spend too much time in close proximity, the system lets the device light up and additionally vibrate, informing the wearer to maintain a proper distance. Here, the use of the AI is determining when to give a warning. For example, when two individuals pass each other in a hallway, there is a low chance of them interacting and posing as a risk. Here, the system can recognize the movement and decide not to send a heavy warning or give a negative score. The latter will be explained further later. Additionally, sub-groups of children and

elderly can be introduced, where the distancing parameter can be respectively less or more strict.

The AI system will keep a log of the behaviour of the bracelet users. When a user is repeatedly or continuously involved in wrong behaviour, the system will send out a warning to the manager's display, who will take action based on the specific situation. For example, the manager may advise the person to leave the party or call for safe transport or in extreme cases, call for additional security personnel. If a general threshold is met with too many violations the AI may decide the party is deemed unsafe and give a suggestion to end it.

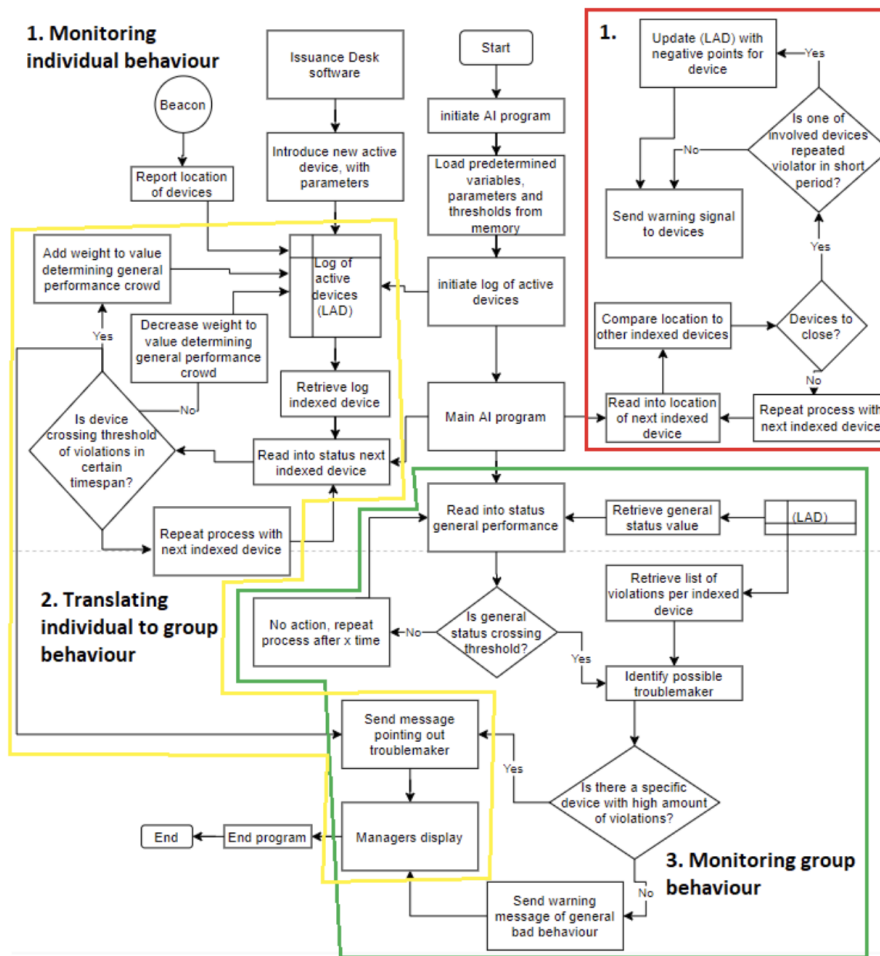
5.1 Flowchart

We have constructed two flowcharts for a basic outline of our designed system. A smaller more generalized flowchart and a bigger more in-depth flowchart explaining the main AI functions. Both can be found in the Appendix. On the next page, the more in-depth flowchart is inserted to elaborate on three processes of the system.

The first process shown in red allows for monitoring individual behaviour and will iterate over every active device comparing its distance to the other active devices. When two devices are found being too close to one another a warning is sent to the wearers and the system log of both devices is updated with a negative score.

The second process shown in yellow allows for the translation of individual behaviour to group behaviour. The AI system will iterate over the potential negative scoring of all active devices. When a negative scoring is encountered the AI system analyzes this score in conjunction with the timespan of the scoring. The general scoring is negatively updated when an individual negative scoring is encountered. Vice versa the general scoring is positively updated when a positive individual scoring is encountered. If the individual scoring is too negative a warning is sent to the manager's display.

The third process shown in green allows for monitoring the group behaviour and will periodically check the group scoring. When this is deemed too negative the AI system will check for possible individuals causing the most trouble. If the negative score cannot be attributed to an individual, the system will conclude that the overall behaviour is at fault and send a warning to the manager's display.



5.2 Technology Background

For an AI-based system to work, the basic need is for the system to have data from its surroundings on which to work. In our case, the data about the distance between two individuals is generated from Bluetooth Low Energy (BLE), also referred to as Bluetooth 4.0, based wristbands. BLE is the power and application friendly version of Classic Bluetooth that was built for the Internet of Things (IoT). In this technology, unlike Classic Bluetooth systems, there is no need to pair the Bluetooth devices. The power-efficiency of BLE energy functionality makes it perfect for devices that run for long periods on power sources. The inbuilt protocol in BLE called PXP (Proximity Profile) lets the proximity sensor detect close contact range which can be predetermined and set. It is slightly approximate, not absolute in the calibration of distances but is well suited for our application.

5.3 Operational Requirements

Every project usually has two aspects to it - a technical and an operational aspect. The technical solution is easier to scope, design and implement. But when it comes to operationalizing the solution, the touchpoints at the human interface are the toughest to handle. The main bottleneck is to overcome the issue of how people will respond to the offered solution because people are inherently resistant to change. This is why the operational aspect can only be planned to a certain extent because of the great uncertainty that lies with the response on the user end.

In the solution proposed in this paper, the main point of possible failure is the step where the bracelets are handed out to the guests. Will they feel comfortable wearing the bracelet or will they refuse? Imposing a rule on people to wear something may face resistance because of the fear of the unknown. They may take this as a privacy intrusion or feel uncomfortable being monitored. Thus, the operational requirement will be to make the process similar to festival bands by making this wristband part of some fun or loyalty schemes so that people feel comfortable, intrigued or fascinated to use them. Furthermore, the necessity of this device to have bars and places of social gathering re-open, will to many be enough motivation to wear the device. Thus, they will feel incentivised to wear them.

6 Evaluation

Research question. Is it possible to increase people’s compliance with social distancing rules in bars through the introduction of an AI-driven wristband solution?

Hypothesis. A solution which creates a “policed” environment where people can socialize in, for example, a bar, whilst having to abide by the social distancing rule, should ultimately affect people’s willingness to abide by it. This is done through physical stimuli which remind the individual to maintain distance if they are not doing so.

Conditions. Control & Experimental

Independent variable. Intervention, the experimental group makes use of the bracelet, while the control group does not. Both the experimental and control group are put in an environment with the same clear visual and verbal reminders. A COVID-19 warning poster at the front door and a welcoming person to remind of the distancing rule. In the testing environment, crosses on the floor made of tape help to indicate the 1,5 meters. In addition to this, the experimental group is also exposed to the bracelet technology.

Dependent variable. We are trying to isolate the level of adherence to the distancing rule, operationalized in the number of violations made by individuals in the testing environment.

Test subjects. Students and possibly others that frequently visit bars.

Possible confounding variables. The bar employees might not make sure that everyone has a bracelet, resulting in improper input from the AI system. The bar employees might find the system interface unwieldy and ignore signals given by the AI system. People might get frustrated by the bracelets and ignore the signals. The system works, but the users do not enjoy using the bracelets. They find it too cumbersome and prefer not wearing it.

Internal validation. Three main validations for the AI system can be identified:

1. Moment to moment distance tracking and correct bracelet signalling.
2. Decision making at an individual level.
3. Decision making at a group level.

To make sure the AI system is correctly managing input and giving a correct output response based on the situation we need to do a lot of scenario testing. In a physical representation of a bar, we can create a testing environment to train and monitor the AI. With participants wearing the bracelets and acting out different scenarios. In this controlled environment, we can set up multiple scripted situations with a predetermined expected outcome. Comparing the AI response to our predetermined outcomes allows us to validate specific areas of the system.

To validate the different steps respectively:

1. Two people approaching each other and distancing again to validate correct tracking and signalling.
2. Having a static crowd with one person approaching everyone excessively to see if the AI system identifies individual troublemakers.
3. Having a static crowd with multiple people approaching everyone excessively to see if the AI system identifies violating behaviour at a group level.

In scenarios two and three, multiple tests are needed to account for different time spans and different group sizes. The AI system should consider these variables when dealing with potential troublemakers.

External validation. To obtain the external validation for the solution, we set up an experiment with a test and control group. We make use of five real-life bar locations, every bar being used for both a test and a control experiment. For every experiment 30 new students are introduced. In total, we will have 300 randomly selected students randomly spread out over 10 experiments in 5 bar locations. For both the control and test setting the bar will have clear distancing instructions on a poster at the front door combined with a welcoming person giving a verbal reminder to keep sufficient distance. Within the bar, crosses on the floor made of tape help to indicate 1,5 meters. In the testing group in addition to the above, we ask the bar owners to make use of the AI system, introducing the bracelets to the test subjects at the front door. The control group will receive nonfunctional bracelets to minimize potential confounding variables.

Both the control and test group will be allowed to socialize for two hours. Two of our group members will monitor the behaviour and record the number of violations between people. They will tally up every time two people stand too close to each other and at the end compare results to validate their results. A decrease in violations for the test groups compared to their control groups should indicate the added value of the AI system. At the same time, the present group members will monitor the response of people on the system, noting positive or negative comments and other potentially more physical feedback such as irritation or enthusiasm.

At the end of each experiment, a performance review will be held with the manager discussing the performance of the AI system. Additionally, the manager will receive a questionnaire based on a Likert-scale to assess their experience. The manager will only receive the questionnaire after the test group has been

finished. The students will also receive a questionnaire based on a Likert-scale to assess their experience. They will receive either the control or test group questionnaire. Both the student and manager questionnaires can be found in the appendix.

7 Conclusion

7.1 Solution

The proposed solution's Platform Software is designed to be a secure, scalable and flexible backend. The combination of the backend platform in conjunction with the companion frontend of Issuance Desk Software or App makes for a robust use case for maintaining social distancing at a bar. Using AI tools, the system can recognize when an individual is ignoring the rules repeatedly and advise the manager on what to do. If the customers comply, this solution makes for a safe environment in bars for students to socialize, which benefits their mental well-being and the bar's financial status.

7.2 General Conclusion

Choosing to deal with a very up-to-date subject as our project assignment, the COVID-19 pandemic, has created an interesting situation. A lot of scientific research that has been done is only months old, with much research still going on right now. In our everyday lives, the pandemic has resulted in stricter rules even going as far as closing all the bars and restaurants for the upcoming weeks, making our proposed solution even more relevant than before.

7.3 Team Evaluation

The pandemic also has a direct effect on our college lives. Every one of us is working from home and the only communication we have to coordinate our efforts is through Zoom meetings and a WhatsApp group. This has not made it easier to work together efficiently. A lot of work has been done individually which affects the overall cohesion of the project. Considering all this, an important lesson has been in understanding the importance of clear and open communication, giving feedback to each other, and managing a clear division of tasks and planning.

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9 Appendix

9.1 Interviews

Interview one: Interview with Drs. M.M. Andalusi. A psychologist that has an education in CBT(Cognitive Behavioural Therapist). The goal of the interview is to get an idea of human behaviour when confronted with wearing bracelets and getting (negative/positive) feedback from an AI on wrong/right behaviour.

– **What do you think of our concept?**

[NL] Goed idee om te onderzoeken, omdat tegenwoordig moeilijk gehandhaafd kan worden. Mensen willen meer vrijheid maar daardoor ontstaat risico. Zeker in grotere sociale omgeving. Het kan mensen bewuster houden aangezien bekend is dat alcohol het bewustzijn verminderd en daardoor het risico wordt verhoogd. Zonder dat dit de intentie is van de persoon.

[ENG] A good idea to research because it is difficult these days to uphold the rules. People want more freedom but as a result, there is more risk. Especially in bigger social environments. It can make people stay more conscious, knowing that alcohol reduces consciousness and thus increasing risk. Without this being the intention of people.

– **Will people feel ok with wearing the bracelet? Will they refuse?**

[NL] Zoals je bij de meeste regels überhaupt ziet is er dimensie van mensen die graag meewerken tot mensen die het belachelijk vinden. Tegelijkertijd weten we dat als het een voorwaarde is van iets wat we graag willen er ook wel weer bereidheid tegenover kan staan. In die zin is het anders dan de regels die altijd voor iedereen gelden, gezien het hier gaat om een keuze om wel of niet naar een café te gaan.

[ENG] As with most rules there will be a dimension of people ranging from being happy to cooperate with people who will find it ridiculous. At the same time when we know that it is a condition for something that we want there can be more willingness. In that regard it is different than rules that always count for everyone, knowing that here there is a choice between either going to a bar or not.

– **Imposing a rule on people to wear something. Is it similar to festival bands?**

[NL] Fysiek is het dezelfde moeite. Mensen zullen het anders ervaren als er meer negatieve consequenties aan hangen. Maar overeenkomsten zijn het draaggemak. De toegang die het geeft tot de sociale activiteit. Hierin zal het reeds lopende proces rondom corona mogelijk ook meer negatieve gedachten kunnen oproepen; minder vrijheid, meer controle.

[ENG] Physically it takes the same effort. People will experience it differently if more negative consequences are attached to it. Similarities are the ease of wearing it and giving access to the social activity. The ongoing situation of the Coronavirus could create more negative thoughts. Less freedom, more monitoring.

– **How will people act once drunk?**

[NL] Dat zou moeten worden onderzocht, mogelijk via een pilot. Het kan twee kanten op, het hangt van de perceptie van de persoon op dat moment af. Dat hangt weer samen met hoe iemand het in het algemeen ervaart, maar dingen als de hoeveelheid alcohol en sociale druk kunnen daarin ook van invloed zijn.

[ENG] That would have to be researched, possibly via a pilot. It can go two ways, depending on the perception of the person at that moment. That again is related to how someone perceives it in general, but things like the amount of alcohol and social pressure can influence this.

– **Is locking the bracelet with a key at the entrance of the party a good idea?**

[NL] Een slot systeem bij de voordeur zou een goede manier zijn om het dragen van de armband als consequente voorwaarde te houden voor de toegang tot het café.

[ENG] A locked system introduced at the front door would be a good way to create a prerequisite condition for receiving entrance to the bar.

– **Will vibrating and a blinking light be enough of a signal to get attention? Considering being sober and drunk.**

[NL] Dit zou wederom moeten worden onderzocht. Als gemeten kan worden dat de persoon een laag bewustzijn heeft zouden daar ook consequenties aan kunnen worden verbonden. Aangezien als je minder bewust van jezelf en je omgeving bent wordt verwacht dat je ook minder rekening houdt met de anderhalf meter afstand.

[ENG] That would once again have to be researched. If it is possible to measure if a person has a low level of consciousness, there could be consequences added to that. Knowing that when you are less conscious of yourself and your environment, it can be expected that you will be less conscious of the one and a half meter distance.

– **What kind of consequences do you think of?**

[NL] Uit onderzoek weten we dat ons gedrag deels gestuurd wordt door de consequenties die daaraan verbonden zijn. Dit kunnen positieve of negatieve consequenties zijn. Iets wat we als beloning of straf ervaren. Deze twee zijn ook weer in twee op te breken. Waarbij je toe kan voegen of weg kan nemen. In het geval van een café zou je op het gebied van straf kunnen denken aan een boete(iets toevoegen) of ontzegging van toegang evt. voor bepaalde tijd(iets wegnemen). In het geval van belonen zou je kunnen denken aan bijv. een voucher voor bitterballen(toevoegen) of een drankje van de rekening af(iets wegnemen).

[ENG] Research shows that our behaviour is partly piloted by the consequences attached to this behaviour. These can be positive or negative consequences. Something we either perceive as reward or punishment. These two can

be divided into two categories: adding something and taking something away. In the case of the bar, considering a punishment could be a fine (adding something) or denying access to the bar for a certain time (taking something away). Considering a reward could be a voucher for free food (adding something) or a discount on the bill (taking something away).

– **Should we use a positive or negative approach?**

[NL] Positieve bekrachtiging zou een mooie toevoeging kunnen zijn op de huidige handhaving omdat mensen het vooral te horen krijgen als ze er niet aan houden terwijl er ook mensen die zich wel moeite doen om zich er aan te houden maar daar geen feedback op krijgen. Negatieve bekrachtiging is hoe er tot nu toe vooral wordt gehandhaafd en waar je wellicht niet snel aan ontkomt op het moment dat mensen belangrijke regels overtreden. Met name wanneer mensen onder invloed zijn van middelen zoals alcohol werkt mogelijk de positieve bekrachtiging minder goed, omdat mensen onder invloed over het algemeen minder bewuste keuzes maken, impulsiever handelen en consequenties minder overzien.

[ENG] Positive reinforcement could be a nice addition to the current enforcement because people mostly get feedback when they are not keeping to the rules, while people who are making an effort to keep to the rules aren't getting any feedback. Negative reinforcement is mostly how we are enforcing the rules right now and is something that is probably difficult to refrain from when people are breaking important rules. Especially when people are under influence of substances like alcohol, positive reinforcement will be less effective, because people under influence generally make less conscious decisions, act more impulsive and are less able to oversee consequences.

– **Do you have any further advice or thoughts to add?**

[NL] Het zou mooi zijn als jullie een opzet zouden kunnen maken dat vervolgens getest zou kunnen worden en uiteindelijk een goede en veilige combinatie zou kunnen bieden van vrijheid en verantwoordelijkheid. Mij lijkt een mooie idee.

[ENG] It would be nice if you could make a design that can be further tested and eventually be able to offer a good and safe combination of freedom and responsibility. To me, it seems like a good idea.

Interview two: Interview with dr. Mieke Donk, associate professor at the VU

– **Do you think people will be okay with wearing a bracelet like this and how will people react?**

I think people will be okay with wearing such a bracelet, but I don't know how people would react to something like that.

– **Would it be a good idea to use the bracelets at private parties?**

I don't think it will be a good idea to use these bracelets at private parties, because I think people don't want to wear it in their own homes, but if you use it at bars, concerts and that kind of events it will work. Especially when people need to buy these bracelets as their ticket to get in. When they go back home they should return it at the entrance and maybe get a part of their money back.

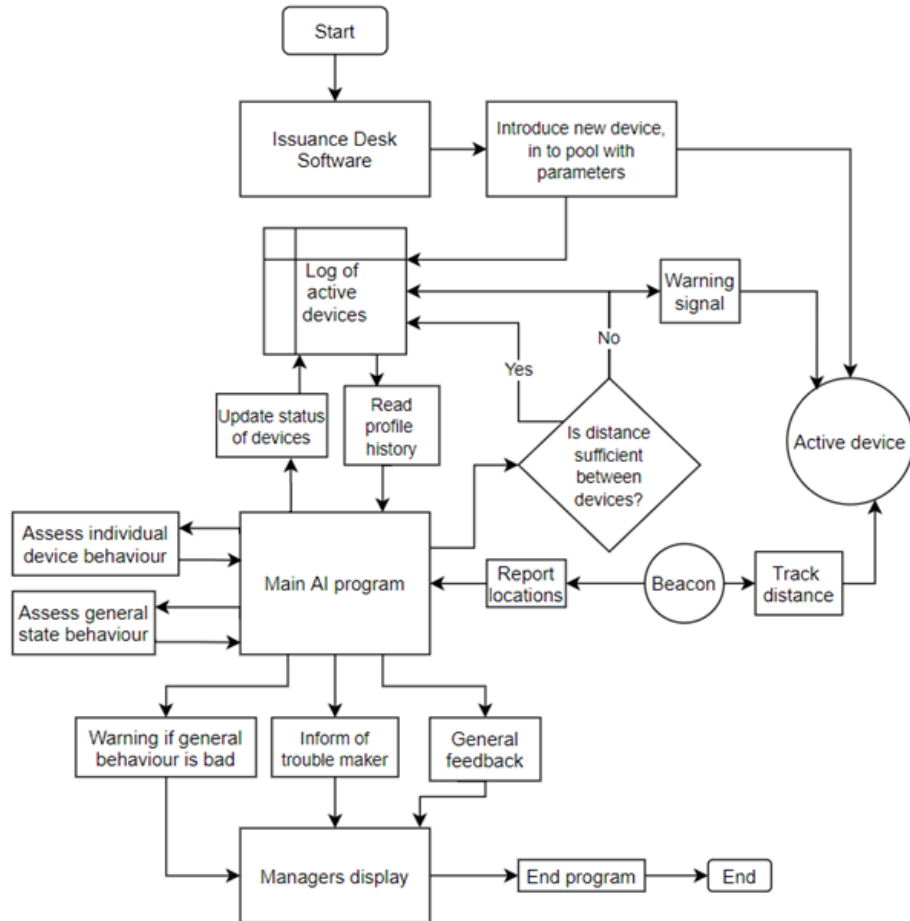
- **As a warning we were thinking of using some kind of noise, but also a light signal, would this be enough as a warning or should add something or remove something?**

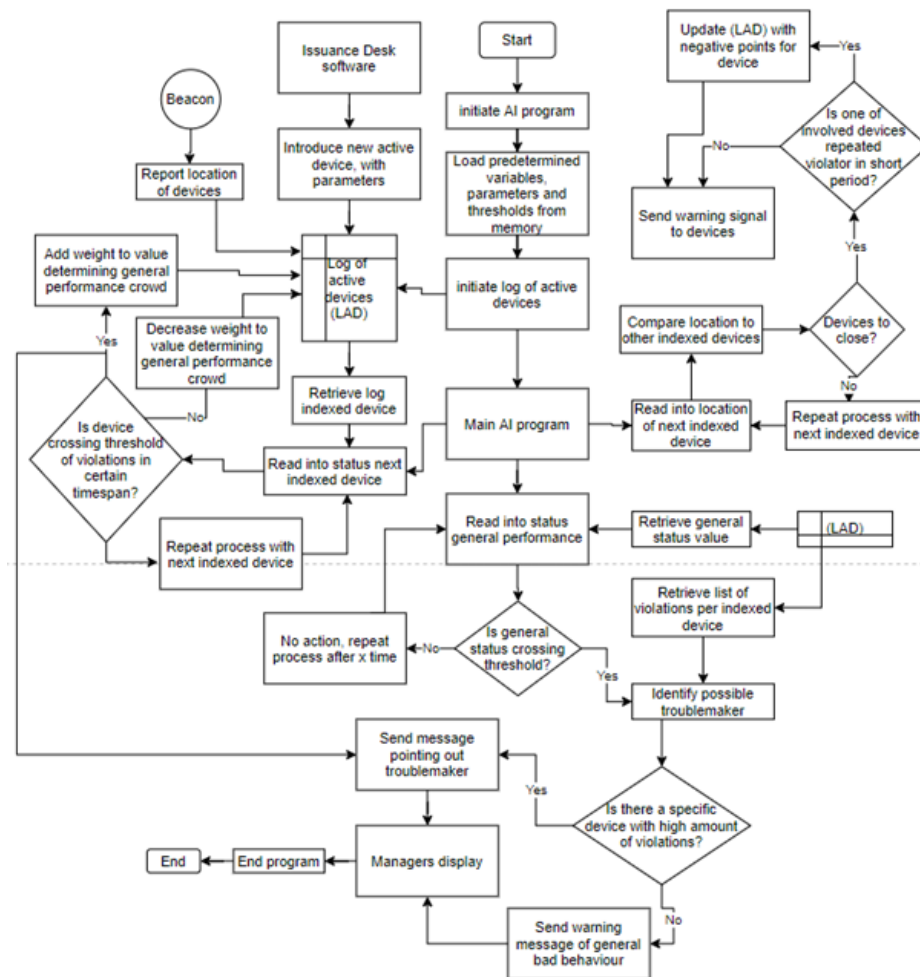
I think if you use a really annoying tone, that would be enough to alert people that they are too close; the light is not necessary in my opinion.

- **Sometimes people get drunk and get too close to each other. How will they act when they have such a bracelet?**

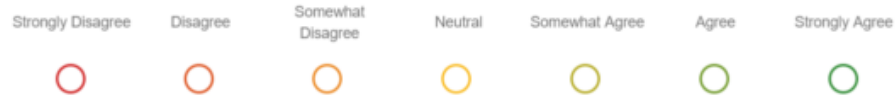
They might try to take it off and throw it away. An idea to prevent this is to find a way that the bracelets can not be taken off on their own. Another idea is to have a really annoying tone when they are too close.

9.2 Flowcharts





9.3 Questionnaires



Questionnaire one Manager questionnaire:

1. The manager's interface of the AI system is easy to use.
2. The manager's interface does not hinder your ability to do your work.
3. The issuance desk interface is easy to use.
4. The issuance desk interface does not hinder your ability to do your work.
5. Introducing the bracelets to customers is an easy process.
6. Introducing the bracelets to customers does not hinder your ability to do your work.
7. The bracelets have a positive influence on the social distancing behaviour of customers.
8. The AI system was helpful in pointing out customers who are not keeping to the distancing rules.
9. The AI system was giving good advice on the general behaviour of the customers, letting you know when the situation was getting out of hand.
10. I would keep using this technology in my bar if possible.

Questionnaire two A Students questionnaire (Control group):

1. I managed to keep an appropriate social distance from other people in the bar.
2. Considering the current Coronavirus, I felt safe socializing with other people.
3. I consider the current Coronavirus a potential threat when going out in a bar setting like this one.

Questionnaire two B Students questionnaire (Test group):

1. I managed to keep an appropriate social distance from other people in the bar.
2. Considering the current Coronavirus, I felt safe socializing with other people.
3. I consider the current Coronavirus a potential threat when going out in a bar setting like this one.
4. Wearing a bracelet did not hinder my enjoyment of socializing at the bar.
5. The signals from the bracelet to keep distance were clear and easy to understand.